Usability Evaluating Methods Cont…

Dr. Sevgi Koyuncu Tunç

Questioning Methods

Questioning Methods aim to learn the feelings and thoughts of the users about a system.

Face-to-face interviews, observing users in the work environment, user tests, and questionnaires are in this group.

User Tests

- User Testing is a way to test software with actual users to see how easy it is to use.
- Users are asked to complete the assigned tasks and are observed by a researcher to see where they have encountered problems and where confused.
- As more users experience similar issues, suggestions are made to overcome these usability issues.

- User tests can uncover which tasks and processes are the most difficult to achieve, which operations are challenging to understand, the user's opinion, feelings, and strains about the system.

User Test

Usability Assessment Process

1 ANALYSIS OF OUTPUTS CAME FROM PREVIOUS TESTS

2 DETERMINING TASKS & PREPARING TEST DATA

3 BUILDING DEMOGRAPHIC INFO FORM AND POST-TEST SURVEY

4 TEST REHEARSAL & IMPROVEMENT OF TEST

5 SPECIFYING TEST USERS & SCHEDULING

6 PREPARING TEST ENVIRONMENT
- Testing video, audio & screen recording
- Testing the screen recorder tool

7 USER TESTS EXECUTION & REPORTING
- Fill in the demographic info form
- Training
- User Test - Task Completion
- Post-test survey
- Opinions

Analysis Of Outputs Came From Previous Tests

- In typical user tests, the tasks assigned to the users are the most frequently used and most essential functions.
- However, to narrow this list, it will be good for the efficiency of the tests to choose the most complex tasks among the frequently used and challenging functions with the Heuristic Walkthrough method before the user test (Albert, Tullis & Tedesco, 2010).

Determining Tasks and Preparing Test Data

- It should be considered that the task scenarios are in line with the real-life workflow provides meaningful results while preparing tasks.
- The relatively short, atomic, and understandable tasks will make users feel comfortable during the user tests.
- Moreover, tasks should be defined to explain the purpose of the assignment without giving clues to the user.

Building Demographic Info Form

- Demographic Info form consists of the following questions that effects the human-computer interaction and usability:
    - Education level
    - Age
    - Computer use level (Insufficient, Partially Sufficient, Sufficient)
    - Internet use level (Insufficient, Partially Sufficient, Sufficient)
    - Do you know about the current software? (Yes/No)

Post-Test Survey

A post-test survey is prepared to handle the opinions of the users on the system and it consists of the questions like in the below list:

1. Finding the information that you are looking for in the system is(Difficult, Normal, Easy)
2. Help and documentation in the system is(Insufficient, Partially Sufficient, Sufficient)
3. Aesthetic features of the system (icons, colour, fonts, etc.) (Bad, Normal, Good)
4. Language of system(Incomprehensible, Partially Understandable, Understandable)
5. The system is(Slow, Normal, Fast)
6. Using and learning the systemis (Difficult, Normal, Easy)
7. Use of the system is (Easy to remember, Partially easy to remember, Not easy to remember)
8. Guideline to prevent the user from making mistakes is(Available, Partially Available, Not Available)
9. The feeling of getting lost in the system is(Experienced, Partly experienced, not experienced)
10. Information messages about the transaction status in the system(Sufficient, Partially Sufficient, Not Sufficient)
11. Error messages in the system in terms of informing and directing (Sufficient, Partially Sufficient, Not Sufficient)
12. Which tasks did you have the most trouble completing during the test? State the reasons.
13. Please indicate if there are any issues you find wrong or missing in the system during the test.

Test Rehearsal and Improvement of Test

No matter how well the user test is designed, it's consistency and operability should be verified from different angles with pilot tests before it is applied. Following issues should be checked in the rehearsal:

- Are the pre-test questions clear?
- Are the tasks well-defined?
- How much time is needed for the training?
- Are there any conditions that technically prevent the tests from being completed?
- Are the final survey questions clear?
- Is the order of the test questions appropriate?

User test should be re-designed according to the results of two consequent pilot tests.

Specifying Test Users and Scheduling

- The number of users to be included in usability testing is a matter of debate among usability experts (Spool, 2001).
- Some studies show no significant difference between the information obtained by testing with large user groups and the information obtained by testing with fewer people since first users reveal most usability problems (Virzi, 1990; 1992).
- The User Test aims to observe the interactions of the people who use the system frequently, less frequently, and for the first time.
- Choosing users from different education levels, ages and genders will enhance the results of the tests.

Preparing Test Environment

- Previous studies show that, the attitudes and behaviours of the users diverge from real life when the user test is performed in an isolated laboratory that is different from the natural work environment.
- And this setting also affects the test accuracy negatively (Brush, Ames & Davis, 2004; Tullis et al.2002; Hertzum, 1999).
- The environment, internet connection speed, amount of noise, amount of light, distractions, and situations that interrupt the work of the user for different reasons should be also included in the test.

User Test Execution and Reporting

User Test is executed with the following steps:

a) Explain the purpose of the test and present the Volunteer Participation Form to the users.

b) Ask users to fill in the demographic information form

c) Short System Training: inform the users about the basic use of system.

d) During the usability test,

- the task completion success rate and task completion times will be recorded manual or by a usability assesment software like Loop11.
- Ask users to express their feelings aloud during the test. The technique of thinking aloud introduced by Karl Duncker enables researchers to gain information about the cognitive processes of the users during the use of the system (Hackos, 1998). However, there may be a risk of mental fatigue and distraction during the test in the subject who is applied thinking aloud. Therefore, the subject should be encouraged to think aloud only at the beginning of the test and not be disturbed during the test process (Nielsen, 2002).
- According to the best practices, the users should not be given clues during the tests unless they request (Barnum, 2011).

e) Present the post-test survey that contains questions about the system at the end of the test.

f) The qualitative and quantitative data obtained from the tests will be brought together and reported.

Think aloud findings

- Kullanılabilirlik testleri sırasında deneklerden sesli düşünmeleri istenmiştir. Kullanım sırasında deneklerin ifade ettikleri duygu ve düşünceleri not edilmiştir. Görevler sırasında en çok kullanılan cümleler aşağıdaki gibidir.

- Kaydedilmedi mi?
- Şimdi ne yapacağım?
- Hangi alanı hatalı girdim, anlayamadım, çok alan var.
- Dosya yaratma ekranında: “Dosya Tanımı ne demek? ne yazacağım? (Dosya adı)
- Önceki ekrana nasıl geri döneceğim?
- İmza planını nasıl ekleyeceğim, bulamıyorum?
- Dosya birleştirmede: 2. Dosyayı nasıl seçiyoruz?
- Evrak aramada: Yarattığım evrakı bulamıyorum.

Findings of User Test – Task Completion

Task No	Task	Avg. Success Rate	Avg. Completion Time (sc)	Max Expected Completion Time (sc)
1	List the records sent to the department. What is the "transaction status" of the 1st record in the list? Receive the record.	100%	156.6	30
2	Create 2 New Folders.	80%	158.8	60
3	Merge the two folders you have created.	20%	148.8	30
4	Query the Unfiled records. File the 1st record in the list.	60%	106.7	40
5	Create a Distribution Plan.	60%	184.3	40
6	Create a new record. Distribute this record to the distribution plan you created in step 5.	50%	221.2	120
7	Find the record from the search page.	80%	77.7	30

Findings of User Test – Post Survey

Post -Test Survey	S 1	S 2	S 3	S 4	S 5	S 6	S 7	S 8	S 9	S 10
Finding the information that you are looking for in the system program (Difficult, Normal, Easy)	D	D	D	D	E	D	D	D	E	N
Help and documentation in system(Insufficient, Partially Sufficient, Sufficient)	I	P	I	I	S	P	P	P	P	P
Aesthetic features of the system program (icons, colour, fonts, etc.) (Bad, Normal, Good)	B	B	N	N	N	N	N	N	B	N
Language of system program is clear (Yes, Partially ,No)	N	P	P	P	P	P	Y	P	P	P
system program is(Slow, Normal, Fast)	S	N	N	N	N	N	S	N	S	N
Using and learning system program, in general, is (Difficult, Normal, Easy)	D	N	D	E	E	N	E	N	N	N
Use of system program is easy to remember (Yes, Partially, No)	N	P	N	Y	P	P	P	N	P	P
Guidelines to prevent the user from making mistakes in the system program (Available, Partially Available, Not Available)	N	N	P	N	P	P	A	N	N	P
Did you feel lost while using the system (Yes, Partially, No)	Y	Y	Y	P	P	Y	P	P	Y	N
Information messages about the transaction status in the system program is (Sufficient, Partially Sufficient, Not Sufficient)	N	P	N	N	P	P	N	P	S	P
Error messages in system program in terms of informing and directing (Sufficient, Partially Sufficient, Not Sufficient)	N	P	S	N	S	S	N	N	N	P

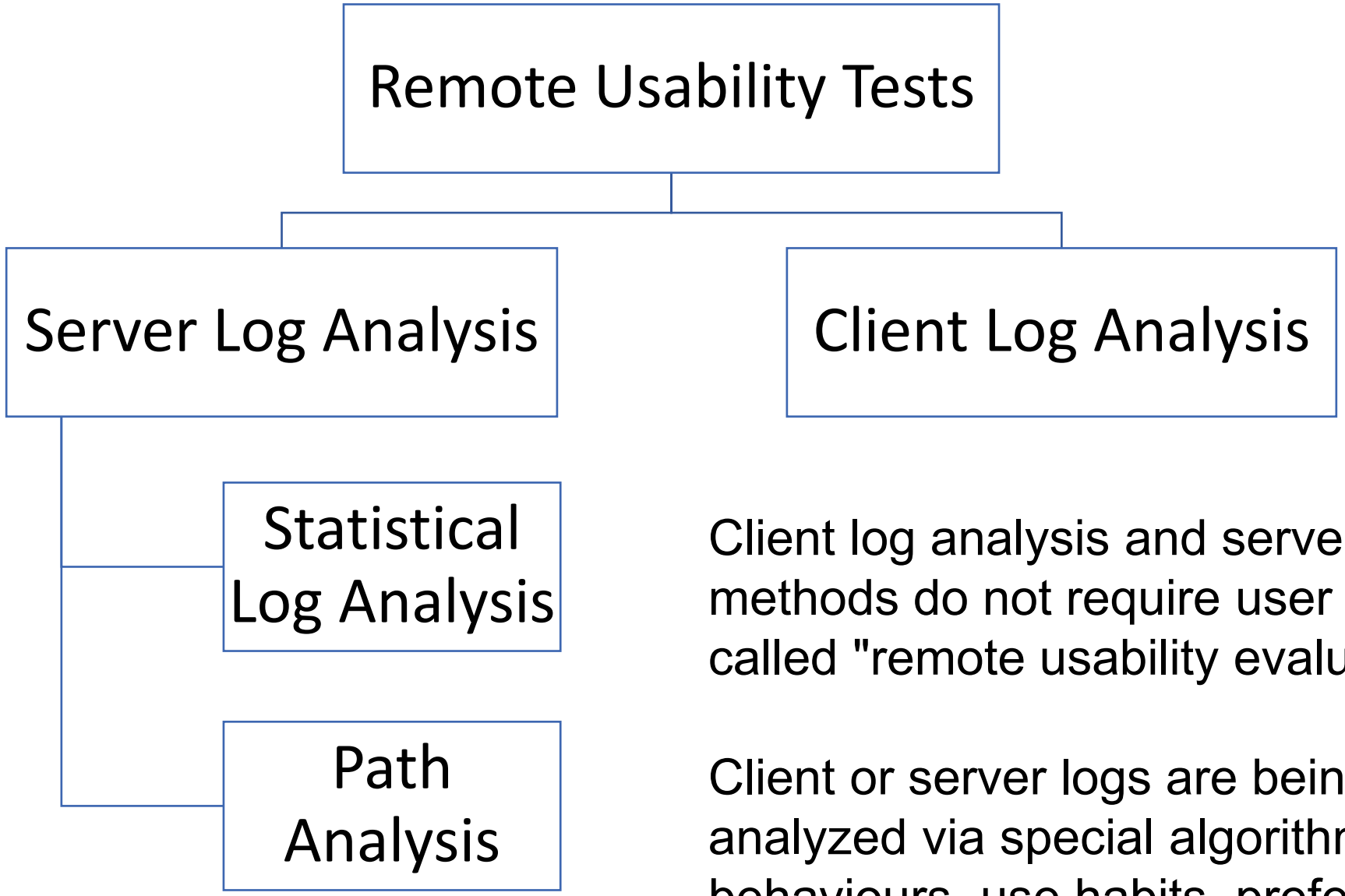
Problems found in User Tests

- Sistem gelen yazıyı teslim almadan dağıtım yapılmasına izin vermektedir.
- Evrak hazırlama ekranlarında tab tuşunun işlevsiz olduğu, veri girişinin zor olduğu gözlemlenmiştir.
- Dağıtım planı hazırlarken listede tüm personel değil ilgili kişiler olmalıdır.
- Çoğu sayfada geri gelme seçeneği yoktur.
- Yardım menüsü kolayca görülebilir bir yerde olmalıdır.
- Sistemde sık sık bağlantı sorunu yaşanmaktadır.
- Dosya yaratma işleminin adımlı bir işlem olduğunu anlaşılmamaktadır. İlk adımı tamamlanıp kaydet butonuna tıklanmakta. kaydetme işleminin başarısız olduğu mesajı dikkat çekmemekte kullanıcı işlemi tamamlandı sanmaktadır.

Problems found in User Tests -2

- Yazışmalar doğru dosyalanamamakta, sınıflandırmada sıkıntı çekilmektedir.
- Yazışma yaratmada gizlilik derecesi ve ivedi durumu alanlarının açılır liste olduğu anlaşılmamakta yazı gibi algılanmaktadır.
- Hata mesajları kullanıcıya yol göstermemektedir.
- Dağıtım planı hazırlama işleminin adımlı bir işlem olduğu kullanıcılar tarafından anlaşılmamıştır.
- Sorgulama ekranları ilk açıldığında boş geldiği için listede kayıt olmadığı sanılmakta, “Sorgula” butonuna basılması gerektiği anlaşılmamaktadır. (Ör: dosyalanmamış evraklar)
- Yazışma ekranı açıldığında gelen boş forma veri girişi yapılmaktadır. Öncesinde “Ekle” butonuna basılması gerektiği kullanıcı tarafından fark edilmemektedir.
- Evrak aramada anahtar kelime ile tüm alanlarda arama yapılması aramayı kolaylaştıracaktır. Filtre alanları anlaşılmamaktadır.

Remote Usability Testing



Client log analysis and server log analysis methods do not require user involvement and are called "remote usability evaluation."

Client or server logs are being pre-processed and analyzed via special algorithms to identify users' behaviours, use habits, preferences, and problems while using the system in their everyday working routine and environment.

Server Log Analysis

- The Server Log Analysis method is used to obtain system usage statistics and user paths to gain insights regarding the usability of web applications and analyse the page access requests in HTTP access logs.

- The server log analysis methods examine the log files recorded during the routine system use of users. Thus, insights can be obtained from both statistical usage data and the paths followed by the users (Bertot et al., 1997).

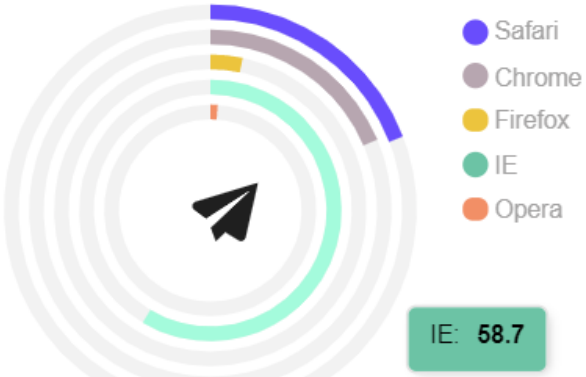
Statistical Log Analysis

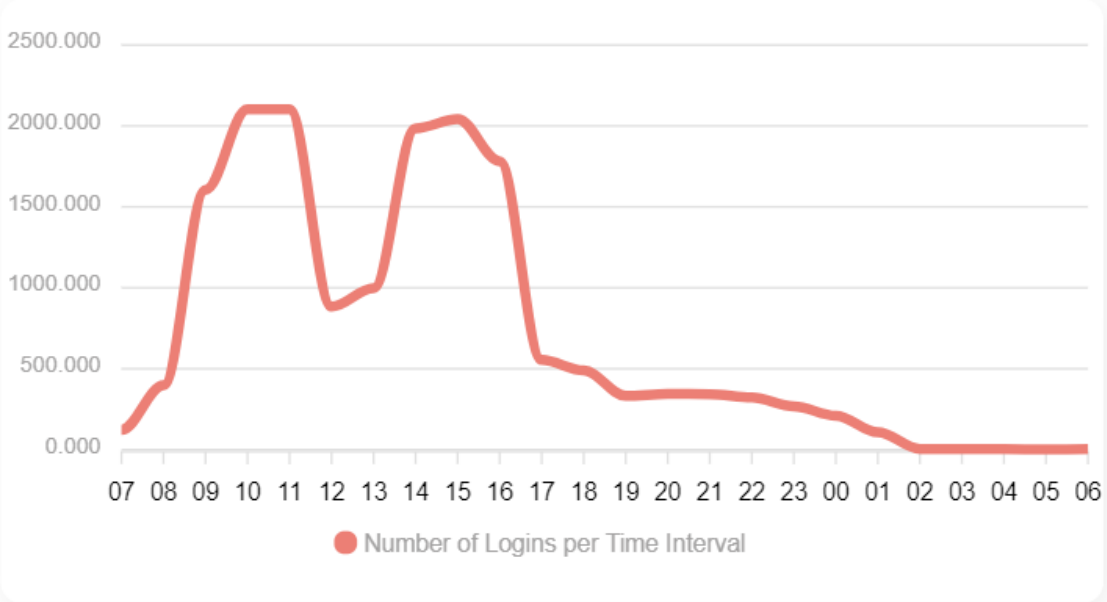
Usability Assessment Process

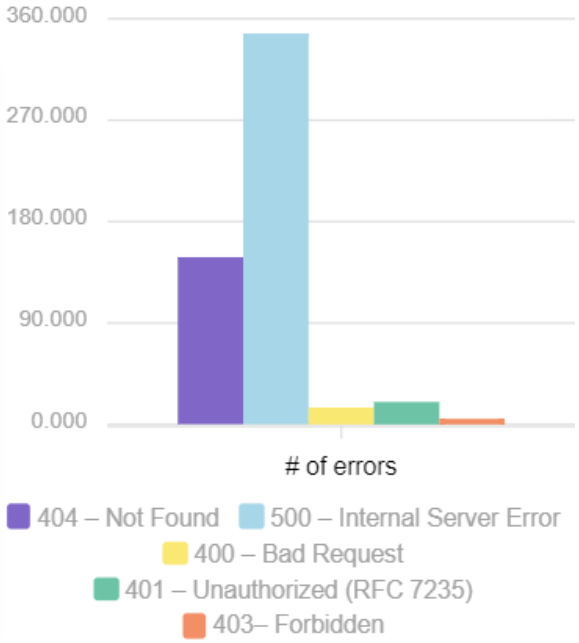
1. ANALYZING & TRANSFERRING LOG FILES TO DATABASE
2. DEVELOPING & VALIDATING SQL QUERIES
3. DEVELOPING AND VALIDATING ALGORITHMS
4. EXECUTION & REPORTING

Statistical Log Analysis Dashboard

# Unique IP	Mobile	Desktop
3416	7%	93%







Top 11 Help Pages	% View
Current Tasks	22,32
Help menu	14,94
Help index	14,21
General Incoming Docs	12,73
Dept. Ongoing Docs	6,09
Dept Ongoing Docs	5,54
Create Document	3,69
List of General Incoming Docs	1,94
General Ongoing Documents	1,85
File merge	1,85
Search Docs	1,66

Top 7 Pages	Avg Time (min)
User Doc. Operations	24,1
Detailed Search	14,7
Create Document	13,13
e-signature	12,53
Current Tasks	10,35
Dept. Ongoing Replies	9,93
Dept. Incoming Docs.	8,82
Search records	7,87

Top 5 Pages	# Requests
Create Document	1.139.674
Current Tasks	441.439
Create a reply doc.	299.659
General Incoming Docs	189.640
Dept Ongoing Replies	128.229
Templates	86.286
Templates2	44.723
Dept Incoming Docs	43.865
Add attachment to a Doc	36.706
Dept Ongoing Docs	21.929

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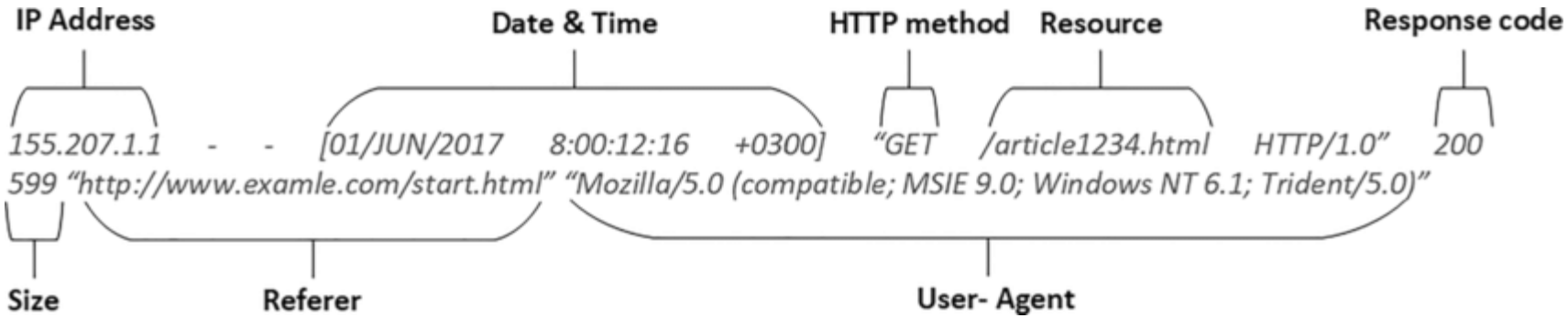
Statistical Log Analysis

- To extract meaningful information about users and the usage from server log records, first of all, log files should be analysed to identify the format and types of the data. A standard “HTTP access log” record consists of the following fields.
  - request date time
  - requested URL
  - the result of the request as a status code
  - length of retrieved page content as bytes
  - remote host: IP address of the us



An “HTTP error log” record consists of the following fields:

- log date time
- severity of the error (error/warn/notice/info etc.)
- IP address of the client
- Error message and the related file’s system path

Database Creation

- After identifying the data types via analysis of log files, data tables should be designed to store a structured version of the log records.
- Server log files are text files with a standard format; thus, special characters separate data fields. Each data field should be parsed based on these special characters and stored in a structured database to be able to query and extract meaningful data.

Statistics from Logs

- # Of Unique IPs
- # Of Mobile Clients' Requests
- Clients' Browser Types
- User Density by Time
- Page view statistics
- Help pages' access statistics
- User-based access statistics
- Average time spent on each page
- Error messages statistics

Findings - Example

Statistical Log Analysis Dashboard

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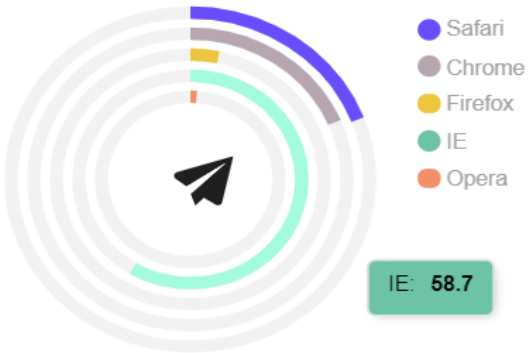

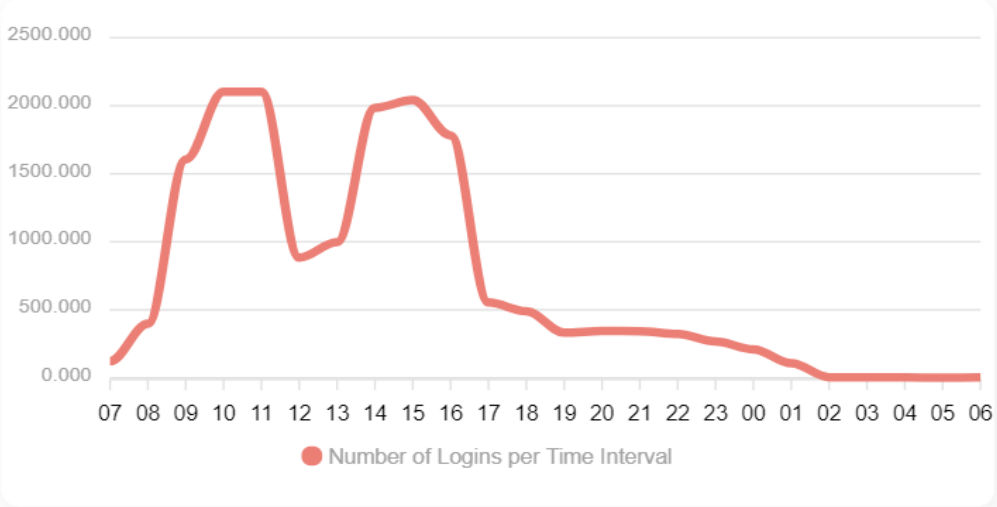

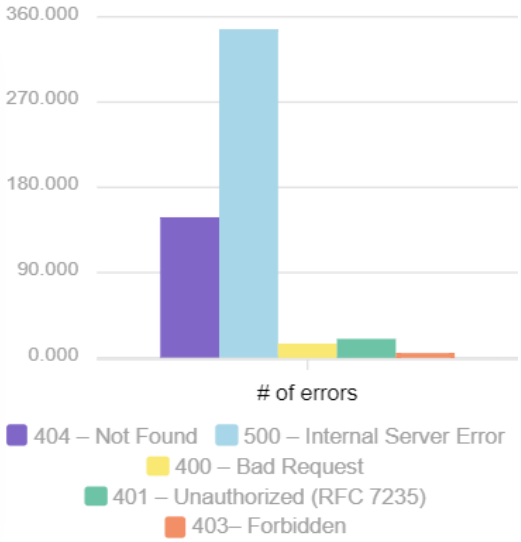

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Statistical Log Analysis

- Software developers can quickly detect abnormal values in page access frequency, error frequency and help page request frequency data obtained from Statistical Log Analysis.
- They can also prevent possible problems by scrutinizing the relevant pages/transactions.
- On the other hand, senior managers and decision-makers can
- take measures to avoid performance degradation during peak hours of use,
- expand browser support in case of different trends in browser preferences,
- take strategic decisions such as extending the scope of mobile applications according to mobile application usage density.

User testing can be applied to investigate the cause of abnormal situations seen in the server logs, and the source of the problems can be determined with user opinion and feedback.

Log Path Analysis

- Log Path Analysis is a web analytics method and studies show that interface problems can be identified by revealing the repetitive patterns created by the users while using the system (Kim, Kang, Kim, 2013).
- Log Path Analysis is a kind of data mining method that requires special algorithms and software that gives an idea about the perception and use of a system among all users.
- The output of the path analysis technique is a ranked list of all sub-paths based on their frequency among all the sessions.
- Another outcome of the path analysis is the pages that are frequently visited together. It is necessary to investigate whether these consecutive pages can be combined in a single page or not to enhance the system's efficiency. In addition, it is possible to improve the web application by adding links between pages which are not related to each other, but which are shared by users.
- Finding Web pages accessed one after the other makes it easier to understand users' activities, trends, and predictions for the next page to visit.

Log Path Analysis

Usability Assessment Process

1. PURIFYING LOG DATA
2. ENCODING UNIQUE URL ADDRESSES
3. SESSION DETECTION
4. EXTRACTING URL SEQUENCE OF EACH SESSION
5. EXTRACTING ALL SUB URL SEQUENCES OF SESSIONS
6. CALCULATING THE FREQUENCY OF EACH URL SEQUENCE
7. ANALYSIS OF THE RESULTS & REPORTING

Purifying data

- Server logs contain web page requests and also some image files, style files and script files that are being transferred during a web page request.

- These irrelevant records should be removed from the log data via SQL queries to purify the database and focus on the related data.

Encoding web page URLs

- A URL path consists of multiple URLs (URL array) visited during a web session.
- Web page URLs can be 2048 characters which makes readability of URL arrays very hard.
- So, each page has given a code to enhance the readability and the representation of the URL arrays.

Identifying user sessions

- Each user spends some time visiting a website with an IP address, and each visit is called a session.
- Sessions can be distinguished by time-oriented or navigation-oriented approaches.
- For example, session start can be determined according to the total duration of the session, the time spent on the page, or the page source like the login page(Spiliopoulou, 2001).
  - session duration =15 minutes,
  - first page call time =t0,
  - next page call time = t1,
  - t1- t0 < 15

Extracting sub-URL sequences and frequency of each unique URL sequence

After session detection, URL sequences are ready to analyse.
Not only the entire URL arrays but all the sub-arrays of URLs can be extracted and analysed to see the behavior of users at micro-levels.
To achieve this, all unique (4 to 20 length) URL sequences in the session table should be extracted and the total number of occurrences of URL sequences in all sessions should be calculated and stored in a table.

For example, if the URL sequence is "G, İ, Y, S, Y, FK" in a session record, the following URL sequences will be extracted from this session record:
Table 1: Sub-paths of "G -> İ -> Y -> S -> Y -> FK" Path

4-page	5-page	6-page
G -> İ -> Y -> S	G -> İ -> Y -> S -> Y	G -> İ -> Y -> S -> Y -> FK
İ -> Y -> S -> Y	İ -> Y -> S -> Y -> FK	
Y -> S -> Y -> FK		

Analysis of URL sequences via SQL queries

- The most frequent and least frequent URL sequences should be identified via SQL queries.
- Also, the pages that are usually accessed together should be revealed and investigated.

Log Path Analysis Insights

- In the Log Path Analysis technique, all users' usage, tendency, and behaviour are investigated, unlike the user test that covers small user groups.
- The path analysis technique makes it possible to draw meaningful results from "big data" with special algorithms that detect the sessions accurately.
- As a result, it can help system administrators
  - To prioritize problems,
  - To increase the effects of the changes on the design or functionality over time,
  - and to discover the pages frequently visited together to optimize the workflows and user interface design.

Client Log Analysis

Client Logs

- In the client log analysis method, all the mouse and keyboard activities of the users are recorded in the database via a piece of code added to the HTML page.
- Usability of the system can be measured by obtaining information such as task completion rate and time, navigation behaviour, and cost of accessing information.
- In this method, it is crucial to ensure the security of the client data not to violate the users' privacy.

Client Logs

- The most important feature of client logs is that they contain high-depth and detailed data.
- However, this method requires placing a piece of code between the main html elements of the website.
- The sheer amount of data available on the client side can be confusing as to what to analyze.
- What data is recorded should be determined and limited to serve the purpose of the application or research topic to be tested.
- There are many resources in the literature on what types of data can help measure usability on the client side.

What to log?

- Roaming Behavior; used to get the most used navigation route on the website.
- the most frequently used pages to enter the website
- the most frequently exited pages,
- number of page views,
- Time-based metrics:
    - the average time spent on each page,
    - mouse dwell time on page elements (for indecision/anxiety detection),
    - time to fill in the form fields,
    - click frequency,
    - write speed.
- Mouse positions, click and navigation coordinates,
- Window size change gestures, page scrolling (must be saved with the page vertical position).
- the time the application is running,
- wait/delay time,
- 7 days active user,
- Participation,
- New user registration,
- loyalty,
- task success
- task completion time,
- problem solving (looking at the help page, using the search feature, faltering) time,
- percentage of achieving the goal,
- user-based performance

Heat Maps

- There is a ~85% correlation between eye and hand movements (Chen, Anderson, Sohn, 2001).
- Heat maps show the areas on the page that attract the most attention of the user.
- It has been revealed that users usually focus the mouse on the point where they pay attention, and 35% of users use the mouse as a reading aid.
- As a result, the focus of the mouse points to the point of interest of the user and helps us to see the user's actions such as reading, thinking, navigating, using the menus on the website.

LuckyOrange

- Luckyorange is one of the most comprehensive client-side logging tools. It offers
- heatmap,
- user session video recording,
- job abandonment statistics,
- asking users short questions during use,
- learning their ideas and chatting with users.

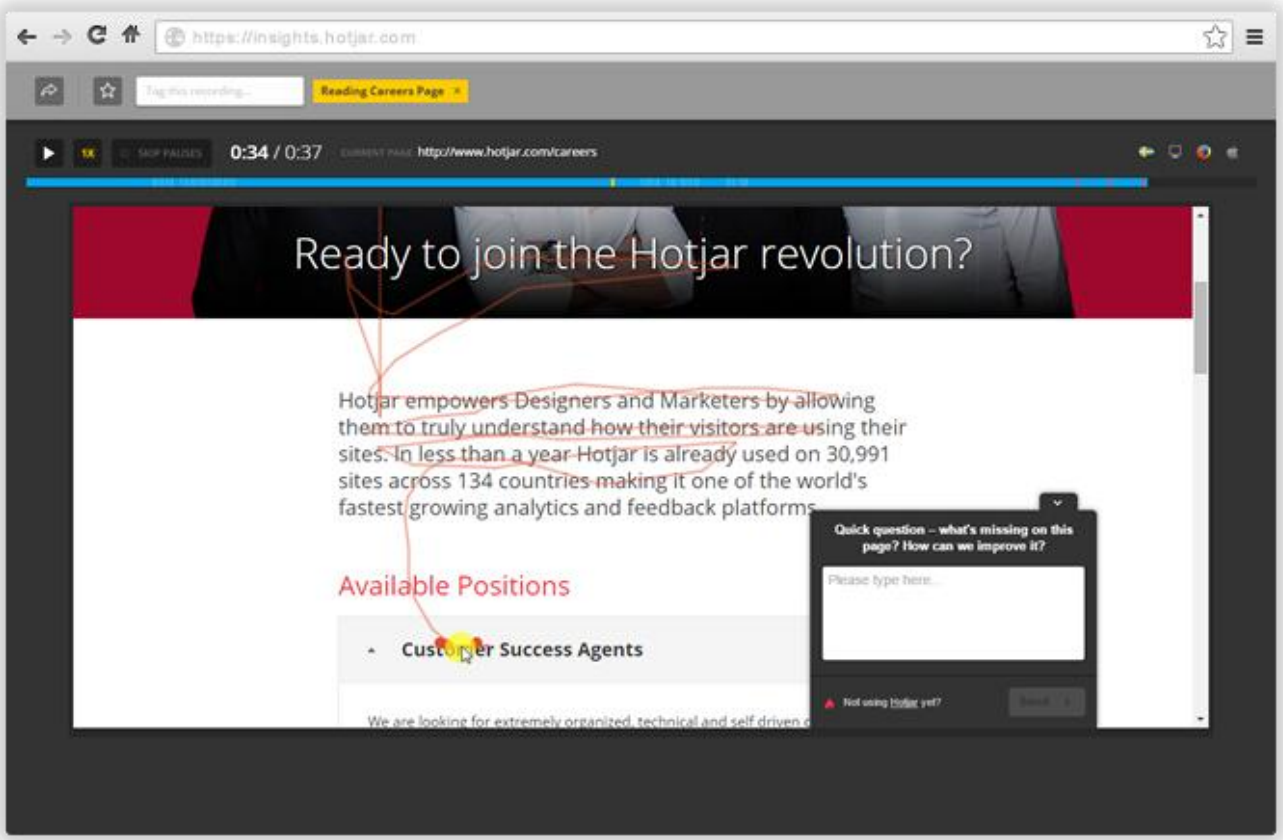
Thus, it is possible to obtain both qualitative and quantitative data.

Hotjar

HotJar generates heat maps and reports using mouse movements and click records.

It also has the feature of preparing surveys for researchers and presenting them to users.

HotJar can report on which step of the task the users leave the site the most, and the form fields that take the longest to fill.



Loop 11

- Loop11 is a usability testing tool that provides automatic usability testing and records user movements and task completion times.
- To use this tool, first an account is opened, then test pages and tasks are set.
- Loop11 generates a link address to be forwarded to potential users and offers different options to deliver this link address to users (showing it as a pop-up on the website, copying the link address and distributing it to users, etc.).
- Users perform the specified tasks using test tools and test data is recorded.
- This tool, which works without adding any code to web pages or web server, can also be used for usability testing and comparison of different websites in the same category.

Formative Tests

Formative tests allow users

- to express their opinions and suggestions,
- to ask questions about the design of the product,
- to optimize and redesign the product based on the feedback

Eye Tracking

- ***Eye Tracking:*** In the Eye Tracking method, the user's eye movements are recorded with special software during the system use. Via this method, predictions can be obtained about the user's functions, intention, and mental state (Cooke, 2006).

The eye tracking method determines where, how long and how many times the user looks, instantaneously and historically.
provides information on where his attention is focused, his intentions, and his mental state.

Usage areas usability tests, market researches,
psychological studies.

Application of Eye Tracking

1. Target Audience Determination and Scenario Creation:
•Identify the target audience for the test.
•Define specific tasks or scenarios that users will be asked to complete.
2. Test:
•Record user behavior while they perform the specified tasks.
•Capture mouse movements, eye movements, sounds, and images.
3. Analysis and Reporting:
•Analyze the recorded data using tools like heatmaps, gaze plots, and user videos.
•Identify patterns, areas of interest, and potential pain points.
•Compile findings into a comprehensive report, including success rates, survey results, and key insights.
4. Implementation:
•Use the insights from the report to make design improvements and optimizations.
•Implement design changes and test their effectiveness.

Findings of Eye Tracking

Temperature Maps: Visual representations of user gaze patterns, highlighting areas of interest and dwell time on specific elements.
User Videos: Recordings of users completing tasks, including their verbal thoughts and facial expressions.
Roadmaps: Visualizations of user journeys, showing how many users followed specific paths and the elements they interacted with.
Mouse Movements: Data on user mouse clicks, including timing, location, and frequency.
Time Statistics: Metrics related to user interaction time, such as time spent on pages, task completion time, and link click duration.

Comparing Usability Testing Methods

Heuristic Walkthrough

Outputs / Data Types / Insights	Heuristic Walkthrough
Usability design flaws	✓
Design improvement suggestions	✓
Processes designed against the workflow	✓
Relevance & directivity of error messages	✓
Efficiency of process steps	✓
Issues with data availability and presentation	✓
Legibility	✓
Consistency between visual and text elements	✓
Overall integrity and consistency	✓
Real user's implementation of business rules	
User's feedback	
how the user uses the computer (short-cuts, menus, etc) and what conveniences he wants.	
Detection of functions that the user is not aware of	
Performance with the user's network connection	
Items not understood by the user	
Think-aloud data	
Software usage frequency	
Usage intensity per hour	
Help page usage	
Error statistics	
Browser preferences	
Error statistics	
Average time spent on the pages	
Most followed paths by users	
The paths that users followed the least	
The ability to reveal anomalies by comparing the predicted path with the most followed paths.	
Facilitate the navigation of the web application by adding links between the most visited pages together	
Predicting user's future circulation by automatically classifying user circulation.	
Users' web page visiting habits, frequency and behavior	

Statistical Log Analysis

Outputs / Data Types / Insights	Statistical Log Analysis
Usability design flaws	
Design improvement suggestions	
Processes designed against the workflow	
Relevance & directivity of error messages	
Efficiency of process steps	
Issues with data availability and presentation	
Legibility	
Consistency between visual and text elements	
Overall integrity and consistency	
Real user's implementation of business rules	
User's feedback	
how the user uses the computer (short-cuts, menus, etc) and what conveniences he wants.	
Detection of functions that the user is not aware of	
Performance with the user's network connection	
Items not understood by the user	
Think-aloud data	
Software usage frequency	✓
Usage intensity per hour	✓
Help page usage	✓
Error statistics	✓
Browser preferences	✓
Error statistics	✓
Average time spent on the pages	✓
Most followed paths by users	
The paths that users followed the least	
The ability to reveal anomalies by comparing the predicted path with the most followed paths.	
Facilitate the navigation of the web application by adding links between the most visited pages together	
Predicting user's future circulation by automatically classifying user circulation	
Users' web page visiting habits, frequency and behavior	
At which step the software was abandoned the most	

Server Log Path Analysis

Outputs / Data Types / Insights	Server Log Analysis
Usability design flaws	
Design improvement suggestions	
Processes designed against the workflow	
Relevance & directivity of error messages	
Efficiency of process steps	✓
Issues with data availability and presentation	
Legibility	
Consistency between visual and text elements	
Overall integrity and consistency	
Real user's implementation of business rules	
User's feedback	
how the user uses the computer (short-cuts, menus, etc) and what conveniences he wants.	
Detection of functions that the user is not aware of	✓
Performance with the user's network connection	
Items not understood by the user	
Think-aloud data	
Software usage frequency	
Usage intensity per hour	
Help page usage	✓
Error statistics	
Browser preferences	
Error statistics	
Average time spent on the pages	
Most followed paths by users	✓
The paths that users followed the least	✓
The ability to reveal anomalies by comparing the predicted path with the most followed paths.	✓
Facilitate the navigation of the web application by adding links between the most visited pages together	✓
Predicting user's future circulation by automatically classifying user circulation	✓
Users' web page visiting habits, frequency and behavior	✓
At which step the software was abandoned the most	✓

Client Log Analysis

Outputs / Data Types / Insights	Client Log Analysis
Usability design flaws	
Design improvement suggestions	
Processes designed against the workflow	
Relevance & directivity of error messages	✓
Efficiency of process steps	✓
Issues with data availability and presentation	
Legibility	
Consistency between visual and text elements	
Overall integrity and consistency	
Real user's implementation of business rules	✓
User's feedback	
how the user uses the computer (short-cuts, menus, etc) and what conveniences he wants.	
Detection of functions that the user is not aware of	✓
Performance with the user's network connection	✓
Items not understood by the user	✓
Think-aloud data	
Software usage frequency	
Usage intensity per hour	
Help page usage	
Error statistics	
Browser preferences	
Error statistics	
Average time spent on the pages	✓
Most followed paths by users	✓
The paths that users followed the least	✓
The ability to reveal anomalies by comparing the predicted path with the most followed paths.	✓
Facilitate the navigation of the web application by adding links between the most visited pages together	✓
Predicting user's future circulation by automatically classifying user circulation	✓
Users' web page visiting habits, frequency and behavior	✓
At which step the software was abandoned the most	✓

User Test

Outputs / Data Types / Insights	User Test
Usability design flaws	✓
Design improvement suggestions	
Processes designed against the workflow	✓
Relevance & directivity of error messages	✓
Efficiency of process steps	✓
Issues with data availability and presentation	✓
Legibility	✓
Consistency between visual and text elements	✓
Overall integrity and consistency	
Real user's implementation of business rules	✓
User's feedback	✓
How the user uses the computer (short-cuts, menus, etc) and what conveniences he wants.	✓
Detection of functions that the user is not aware of	✓
Performance with the user's network connection	✓
Items not understood by the user	✓
Think-aloud data	✓
Software usage frequency	
Usage intensity per hour	
Help page usage	
Error statistics	
Browser preferences	
Error statistics	
Average time spent on the pages	
Most followed paths by users	
The paths that users followed the least	
The ability to reveal anomalies by comparing the predicted path with the most followed paths.	
Facilitate the navigation of the web application by adding links between the most visited pages together	
Predicting user's future circulation by automatically classifying user circulation	
Users' web page visiting habits, frequency and behavior	
At which step the software was abandoned the most	

Eye Tracking

Outputs / Data Types / Insights	Eye Tracking
Usability design flaws	✓
Design improvement suggestions	
Processes designed against the workflow	✓
Relevance & directivity of error messages	✓
Efficiency of process steps	✓
Issues with data availability and presentation	✓
Legibility	✓
Consistency between visual and text elements	✓
Overall integrity and consistency	
Real user's implementation of business rules	✓
User's feedback	✓
How the user uses the computer (short-cuts, menus, etc) and what conveniences he wants.	
Detection of functions that the user is not aware of	✓
Performance with the user's network connection	
Items not understood by the user	✓
Think-aloud data	✓
Software usage frequency	
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At which step the software was abandoned the most	

Why we should assess the usability of the software systems?

- The most critical factor that affects the business efficiency in the software products is the user and the system's usability.
- Therefore, the usability of software products should be assessed periodically both in the software development life cycle and during the usage of the system.
- So, problems and defects can be identified, and the software can be optimized.
- The usability test methods
  - can assess the usability of the systems and reveal the usability problems,
  - can provide information for supporting strategic decisions,
  - can help monitor the user trends of the systems.

- All this information serves the purpose of providing better service to users and to optimize the system’s usability continuously.

USABILITY ASSESSMENT MODEL

HEURISTIC WALKTHROUGH

- problematic visual design
- issues on workflows
- weak data presentation
- unclear messages, help pages
- poor navigation design
- error tolerance/management

ISSUE REPORT

- suggestions on design problems
- workflows contrary to domain standards
- the propriety and directivity of error messages
- inefficient workflows/steps
- issues & solutions abaout data presentation
- legibility issues
- consistency between visual and text items
- general integrity issues

STATISTICAL LOG ANALYSIS

- software usage frequency
- usage density per hour
- help page access
- error statistics
- browser preferences
- average time spent on pages

STRATEGY DETERMINATION VIA USAGE

- deciding maintenance schedule by the usage density per hour
- prioritizing the incidents pointed by error statistics
- optimizing performance based on intensity of use
- extending/improving support for new browsers by the user browser preferences
- improving/prioritizing mobile services by the mobile access tendency

LOG PATH ANALYSIS

- frequently followed paths
- under-taken paths
- page groups accessed together
- change in popular paths over time
- pages that users leave the system

ANOMALIES & INSIGHTS ON USER TRENDS

- observing the difference between the expected path and frequently followed path
- monitoring how design changes affect user behavior
- measuring changes in user behavior over time

USER TEST

- user's feelings and thoughts on the system
- how the user uses the system
- complications in accessing information
- functions that the user is not aware of
- system execution in the user's network connection
- items that are not understood by the user
- processes designed against workflow
- the appropriateness and directivity of error messages
- the efficiency of the process steps
- problems with data usability and presentation
- legibility
- integrity and consistency issues

USABILITY ENHANCEMENTS

- Reconstructing workflow steps
- Improvement in interface visual design
- Improving data presentation
- Improvement in error/ warning/ information messages
- Help documents improvement
- Redesigning navigation mechanism
- Preventing transactions against domain standards